

1.Tamper-Evidence in Blockchain

Before Tampering:

The blockchain contains original data stored inside blocks. Each block has a unique hash that depends on its data and the previous block's hash. If the data is unchanged, the hash remains the same and the blockchain is valid.

After Tampering:

If someone modifies the data inside a block, the block's hash changes. This breaks the connection with the next block because the previous hash no longer matches. The blockchain validation process detects this change and shows that the data has been tampered with.

Before Tampering

```
PS C:\Users\shimr\Desktop\toy-blockchain> go run ./cmd
-----
Block mined!
Nonce: 1313
Hash : 000f870bb080b03a8a645393f9aefec5d35f9720cb4af7ea1654900bffc8218
Time : 1.0699ms
-----
Block mined!
Nonce: 14066
Hash : 00007ca5f96bf5e8d4681678948d4e6e6c0e851b3492711edea5d957ead1438e
Time : 11.5587ms
-----
Blockchain created!

Index: 0
Timestamp: 0
Transactions: []
Previous Hash: 0000000000000000000000000000000000000000000000000000000000000000
Hash: c6adee0cd75c8a8a88423ed4adec552a4ecc35253aaac7b85e1fc71376ef5d73
-----
Index: 1
Timestamp: 1783590588
Transactions: [{Alice Bob 50}]
Previous Hash: c6adee0cd75c8a8a88423ed4adec552a4ecc35253aaac7b85e1fc71376ef5d73
Hash: 000f870bb080b03a8a645393f9aefec5d35f9720cb4af7ea1654900bffc8218
-----
Index: 2
Timestamp: 1783590588
Transactions: [{Bob Charlie 20}]
Previous Hash: 000f870bb080b03a8a645393f9aefec5d35f9720cb4af7ea1654900bffc8218
Hash: 00007ca5f96bf5e8d4681678948d4e6e6c0e851b3492711edea5d957ead1438e
-----
insufficient balance
amount must be greater than zero

Current Balances
Charlie : 20.00
Alice : 50.00
Bob : 30.00

Current Balances
Charlie : 20.00
Alice : 50.00
Bob : 30.00
Blockchain is valid ✓
PS C:\Users\shimr\Desktop\toy-blockchain> |
```

Figure 1:Blockchain is valid.

After Tampering

```
insufficient balance
amount must be greater than zero _
Bob : 30.00
Blockchain is valid ✓
PS C:\Users\shimr\Desktop\toy-blockchain> go run ./cmd
-----
Block mined!
Nonce: 18036
Hash : 00015b585d9fd223cacfb778dade8f86fcd0bd08646cbba1a8c19a31ad6cb2af
Time : 13.4747ms
-----
Block mined!
Nonce: 1256
Hash : 000addfb00d67ef767430e3d15c76dc0668342ab428c9cf85c1ed022d3a946b
Time : 612.8µs
-----
Blockchain created!

Index: 0
Timestamp: 0
Transactions: []
Previous Hash: 0000000000000000000000000000000000000000000000000000000000000000
Hash: c6adee0cd75c8a8a88423ed4adec552a4ecc35253aaac7b85e1fc71376ef5d73
-----
Index: 1
Timestamp: 1783590694
Transactions: [{Alice Bob 50}]
Previous Hash: c6adee0cd75c8a8a88423ed4adec552a4ecc35253aaac7b85e1fc71376ef5d73
Hash: 00015b585d9fd223cacfb778dade8f86fcd0bd08646cbba1a8c19a31ad6cb2af
-----
Index: 2
Timestamp: 1783590694
Transactions: [{Bob Charlie 20}]
Previous Hash: 00015b585d9fd223cacfb778dade8f86fcd0bd08646cbba1a8c19a31ad6cb2af
Hash: 0000addfb00d67ef767430e3d15c76dc0668342ab428c9cf85c1ed022d3a946b
-----
insufficient balance
amount must be greater than zero

Current Balances
Bob : 30.00
Charlie : 20.00
Alice : 50.00

Current Balances
Alice : 50.00
Bob : 30.00
Charlie : 20.00
Validation Failed ✗ Invalid hash at block 1
PS C:\Users\shimr\Desktop\toy-blockchain> |
```

Figure 2:Validation Failed

2. Difficulty Versus Effort

Results

Difficulty	Nonce Attempts	Mining Time
2	196	511 μ s
3	6,933	6.0307 ms
4	52,316	34.1005 ms

Analysis

The results clearly show that mining becomes significantly more expensive as the difficulty increases.

At difficulty level 2, only 196 nonce values were tested before a valid hash beginning with two zeros was found. Increasing the difficulty to 3 required 6,933 attempts, while difficulty 4 required more than 52,000 attempts.

Mining time followed the same trend, increasing from approximately 0.5 milliseconds to over 34 milliseconds.

This growth is **not linear**. Each additional required leading zero greatly reduces the probability of finding a valid hash, meaning many more hashes must be tested before success. Mining is therefore closer to exponential growth than linear growth, although the exact number of attempts varies because SHA-256 behaves like a random function.

3. Design Write-Up

Hashing Scheme

Each block hash is generated using the SHA-256 hashing algorithm.

The following fields are concatenated in a fixed order before hashing:

1. Block Index
2. Timestamp
3. Transactions
4. Previous Hash
5. Nonce

Using a fixed ordering ensures that identical block contents always produce the same hash. If any field changes, even by one character, SHA-256 generates a completely different output due to the avalanche effect.

The nonce is repeatedly changed during mining until the generated hash satisfies the required mining difficulty.

Validation Process

The blockchain validation process performs three main checks for every block:

1. Hash Verification

The hash is recalculated using the block contents and compared with the stored hash.

2. Previous Hash Verification

Each block stores the hash of the previous block. Validation checks that this value exactly matches the previous block's hash.

3. Proof-of-Work Verification

The validator checks that every mined block's hash begins with the required number of leading zeros.

Together, these checks guarantee the integrity of the entire blockchain. If any block is modified, at least one of these checks fails, causing the blockchain to be rejected.

4. Discussion Questions

1. How does the previous-hash link make tampering with an old block impractical in a real blockchain?

In this toy blockchain, modifying an old block is easy because the blockchain exists on only one computer. An attacker could theoretically recalculate all subsequent blocks without interference.

In a real blockchain, every block contains the previous block's hash, forming a chain of cryptographic links. Changing an old block changes its hash, which invalidates every subsequent block. To successfully tamper with the blockchain, an attacker would need to re-mine every following block while simultaneously producing a chain longer than the honest network. For large public blockchains such as Bitcoin, this would require enormous computing power and is considered computationally impractical.

2. Alternative to Proof-of-Work

One alternative consensus mechanism is **Proof-of-Stake (PoS)**.

Advantage

Proof-of-Stake consumes significantly less electricity because validators are chosen based on the amount of cryptocurrency they stake rather than performing computationally expensive mining.

Drawback

Validators with large amounts of cryptocurrency may gain greater influence over the network, potentially leading to centralization if not carefully designed.

3. Three Differences Between This Toy Blockchain and a Production Blockchain

Difference 1 – Network Consensus

This project runs on a single computer and has no peer-to-peer network. Production blockchains require many independent nodes to agree on the blockchain state through a consensus protocol.

Difference 2 – Digital Signatures

Transactions in this project are not cryptographically signed. Real blockchains use public-key cryptography so that only the owner of an account can authorize spending.

Difference 3 – Merkle Trees

This blockchain stores transactions directly inside each block. Production blockchains use Merkle trees to efficiently verify that individual transactions belong to a block without downloading every transaction.